

Data Engineer 1

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Outline

1. Data overview

1.1. Structure, Unstructure, and Semi-structure

2. Data format

2.1. Data types (CSV, JSON, Parquet, AVRO)

2.2. Exercise 1

3. ETL (Extract, Transform, Load)

4. Practice

4.1. Prefect

1. Data overview

1. Structured data

- Typically categorized as quantitative data
- Developed by IBM in 1974, structured query language (SQL) is the programming language used to manage structured data
- By using a relational database, business users can quickly input, search and manipulate structured data
- Highly organized and easily decipherable (readable) by machine learning algorithms

Tools & Use cases

Tools

- SQLite: A self-contained, serverless, zero-configuration, transactional relational database engine
- MySQL: Embeds data into mass-deployed software, particularly mission-critical, heavy-load production system
- PostgreSQL: Supports SQL and JSON querying as well as high-tier programming languages (C/C+, Java, Python, etc.)

Use cases

Customer relationship management (CRM) software runs structured data through analytical tools to create datasets that reveal customer behavior patterns and trends

- Hotel and ticket business
- Accounting firms or departments

Pros & Cons

Pros

- Easily used by business users who does not require an in-depth understanding of different types of data and how they function
- There are more tools available for using and analyzing structured data.

Cons

- Limited usage: Data with a predefined structure can only be used for its intended purpose
- Limited storage options: Structured data is generally stored in data storage systems with rigid schemas. Therefore, changes in data requirements necessitate an update of all structured data, which leads to a massive expenditure of time and resources.

2. Unstructured data

- Typically categorized as qualitative data
- Cannot be processed and analyzed via conventional data tools and methods
- Does not have a predefined data model, it is best managed in non-relational (NoSQL) databases
- Another way to manage unstructured data is to use data lakes to preserve it in raw form
- The importance of unstructured data is rapidly increasing. Unstructured data is over 80% of all enterprise data, while 95% of businesses prioritize unstructured data management.

Tools & Use cases

Tools

- AWS S3 (Amazon Simple Storage Service): A massively scalable, highly durable cloud object storage service
- Hadoop: Provides distributed processing of large data sets using simple programming models and no formatting requirements.
- Azure: Enables agile cloud computing for creating and managing apps through Microsoft's data centers.

Use cases

- Data mining: Enables businesses to use unstructured data to identify consumer behavior, product sentiment, and purchasing patterns to better accommodate their customer base.
- Predictive data analytics: Alert businesses of important activity ahead of time so they can properly plan and accordingly adjust to significant market shifts

Pros & Cons

Pros

- Native format: Unstructured data, stored in its native format, remains undefined until needed. Its adaptability increases file formats in the database, which widens the data pool and enables data scientists to prepare and analyze only the data they need.
- Fast accumulation rates: Since there is no need to predefine the data, it can be collected quickly and easily.
- Data lake storage: Allows for massive storage and pay-as-you-use pricing, which cuts costs and eases scalability.

Cons

- Requires expertise: Due to its undefined/non-formatted nature, data science expertise is required to prepare and analyze unstructured data.
- Specialized tools: Specialized tools are required to manipulate unstructured data, which limits product choices for data managers.

Structured VS Unstructured data

While structured data gives a “birds-eye view” of customers, unstructured data provides a deeper understanding of customer behavior and intent. The key areas of difference and their implications:

- **Sources:** Structured data is sourced from GPS sensors, online forms, network logs, web server logs, OLTP systems, etc., whereas unstructured data sources include email messages, word-processing documents, PDF files, etc.
- **Forms:** Structured data consists of numbers and values, whereas unstructured data consists of text files, audio and video files, etc.
- **Models:** Structured data has a predefined data model and is formatted to a set data structure before being placed in data storage (e.g., schema-on-write), whereas unstructured data is stored in its native format and not processed until it is used (e.g., schema-on-read).
- **Storage:** Structured data is stored in tabular formats (e.g., SQL databases in data warehouses). Unstructured data is stored as media files or NoSQL databases in data lakes.
- **Uses:** Structured data is used by data analysts, while unstructured data is used by data scientists.

3. Semi-structured data

- Semi-structured data (e.g., JSON, CSV, XML, Markdown)
- The “bridge” between structured and unstructured data.
- It does not have a predefined data model and is more complex than structured data, yet easier to store than unstructured data.
- Semi-structured data uses “metadata” (e.g., tags and semantic markers) to identify specific data characteristics and scale data into records and preset fields. Metadata ultimately enables semi-structured data to be better cataloged, searched and analyzed than unstructured data.
- Example of metadata usage: An online article displays a headline, a snippet, a featured image, image alt-text, slug, etc., which helps differentiate one piece of web content from similar pieces.

Tools & Use cases

Tools

- MongoDB: Uses flexible documents to process data for cross-platform applications and services.
- Apache Cassandra: A wide-column store, i.e., each row doesn't need to have the same columns
- ElasticSearch: Text search and analytics engine

Use cases

- E-commerce analysis
- Sentiment analysis
- Social listening

Recent developments in AI are driving the future wave of data, which is enhancing business intelligence and advancing industrial innovation.

- Analyze digital communications for compliance: Pattern recognition and email threading analysis software that can search email and chat data for potential noncompliance.
- Track high-volume customer conversations in social media: Text analytics and sentiment analysis that enables monitoring of marketing campaign results and identifying online threats.
- Gain new marketing intelligence: ML analytics tools that can quickly cover massive amounts of data to help businesses analyze customer behavior.

Furthermore, smart and efficient usage of data formats and models can help you with the following:

- Understand customer needs at a deeper level to better serve them
- Create more focused and targeted marketing campaigns
- Track current metrics and create new ones
- Create better product opportunities and offerings
- Reduce operational costs

Snowflake can handle all data types.

Some modern technology can handle all data types

Snowflake (<https://www.snowflake.com/en/>) can handle **all three** types of data in one place

- **Structured:** Standard SQL tables (like your sales data).
- **Semi-Structured:** It has a special **VARIANT** column type. You can dump raw **JSON** into a table and query it using standard SQL without "unpacking" it first.
- **Unstructured:** As of 2026, it has a native **FILE** data type. You can store references to PDFs, images, or audio files and use built-in AI (called **Snowflake Cortex**) to "read" or "summarize" them directly with SQL commands.

Example of data types

Unstructured Data

The university has 5600 students. Shaun (ID Number: 160801), 18 years old Communication study. Linh with ID number 160802, majoring in Accounting and is 20 years old. Ahmed from Psychology study program, 19 years old, ID number 160803.

Semi-Structured Data

```
<University>
  <ID Number="160801">
    <Name="Shaun">
    <Age="18">
    <Program="Communication">
  <ID Number="160802">
    <Name="Linh">
    <Age="20">
    <Program="Accounting">
..... </University>
```

Structured Data

ID	Name	Age	Program
160801	Shaun	18	Communication
160802	Linh	20	Accounting
160803	Ahmed	19	Psychology

..... </University>

Exercise 1: Design and Create JSON file

- Editor online: <https://jsoneditoronline.org/>

2. Data format

2.1 CSV

Comma-separated values (CSV) is a text file format that uses commas to separate values, and newlines to separate records. A CSV file stores tabular data (numbers and text) in plain text, where each line of the file typically represents one data record. Each record consists of the same number of fields, and these are separated by commas in the CSV file. If the field delimiter itself may appear within a field, fields can be surrounded with quotation marks.

Year	Make	Model	Description	Price
1997	Ford	E350	ac, abs, moon	3000.00
1999	Chevy	Venture "Extended Edition"		4900.00
1999	Chevy	Venture "Extended Edition, Very Large"		5000.00
1996	Jeep	Grand Cherokee	MUST SELL! air, moon roof, loaded	4799.00

2.2 JSON

JSON: A lightweight data interchange format. It was designed to be easy to read and write for humans, as well as easy to parse and generate for machines.

```
{ "name": "John", "age": 13, "favoriteColors": ["red", "blue", "green"] }
```

2.3 Parquet

Apache Parquet is an open source, column-oriented data file format designed for efficient data storage and retrieval. It provides high performance compression and encoding schemes to handle complex data in bulk and is supported in many programming languages and analytics tools.

<https://parquet.apache.org/>

2.3 Parquet (Continue)

Columnar database	Row-oriented database
✓ High performance for analytical queries	✗ Inefficient for analytical queries
✓ Efficient storage and compression	✗ Suboptimal storage and compression
✓ Ease of scalability	✗ Complex and expensive to scale
✗ Slower for transactional workloads	✓ Efficient for transactional workloads
✗ Complexity in schema design	✓ Easy to understand and implement
✗ Higher overhead for updates and inserts	✓ Easier modification of data

<https://www.couchbase.com/blog/columnar-store-vs-row-store/>

2.3 Parquet (Continue)

1.2.4 Parquet VS CSV

Dataset	Size on Amazon S3	Query Run Time	Data Scanned	Cost
Data stored as CSV files	1 TB	236 seconds	1.15 TB	\$5.75
Data stored in Apache Parquet Format	130 GB	6.78 seconds	2.51 GB	\$0.01
Savings	87% less when using Parquet	34x faster	99% less data scanned	99.7% savings

<https://www.databricks.com/glossary/what-is-parquet>

2.4 AVRO

Apache Avro™ is a data serialization system.

Avro provides:

- Rich data structures.
- A compact, fast, binary data format.
- A container file, to store persistent data.
- Remote procedure call (RPC).

```
{ "namespace": "example.avro",  
  "type": "record",  
  "name": "User",  
  "fields": [  
    { "name": "name", "type": "string" },  
    { "name": "favorite_number", "type": ["int", "null"] },  
    { "name": "favorite_color", "type": ["string", "null"] }  
  ]  
}
```

<https://avro.apache.org/docs/>

a simple schema example

Exercise 2: Example of data sources

- <https://github.com/ekaratnida/Workshop/tree/main/de/Lab1>

3. ETL (Extract, Transform, Load)

Modern data architecture

Data is scattered across various sources (e.g., CRM systems, transactional databases, log files, and third-party APIs). To make data-driven decisions, organizations need a centralized repository.

ETL (Extract, Transform, Load) is the foundational data integration process used to copy data from disparate sources, modify it into a consistent format, and load it into a destination system, typically a Data Warehouse.

Data Pipeline (ETL)

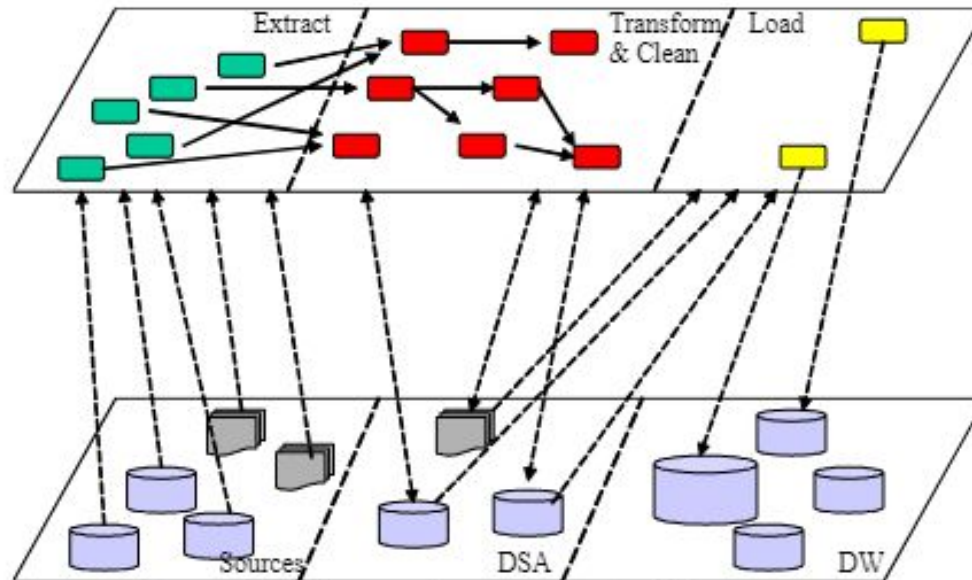


Figure 1. The environment of ETL processes

Vassiliadis et.al., "Conceptual modeling for ETL processes." *Proceedings of the 5th ACM international workshop on Data Warehousing and OLAP*. 2002.

3.1 Extract

Capturing and retrieving raw data from various source systems without impacting their operational performance.

1.1 Sources

Relational databases (MySQL, PostgreSQL),
NoSQL databases (MongoDB),
Files (CSV, JSON, XML),
cloud APIs.

1.2 Extraction Methods

- Full Extraction: High-volume, complete replication of all data. Used during initial setups.
- Incremental/Delta Extraction: Captures only the data that changed since the last run (utilizing timestamps or Change Data Capture (CDC) logs).

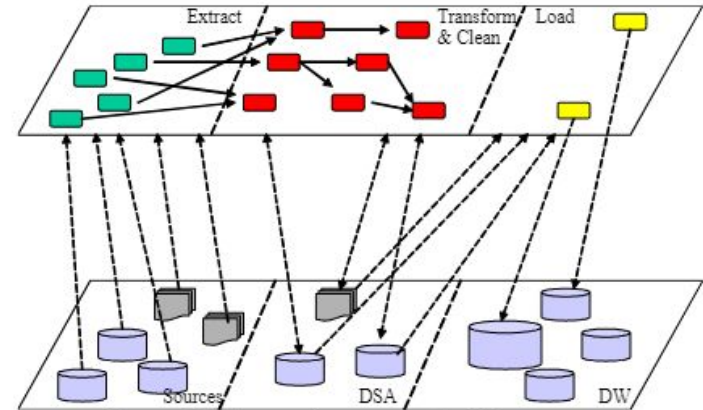


Figure 1. The environment of ETL processes

3.2 Transform

Applies a set of rules and functions to clean and structure the data.

Key Operations

- Cleaning: Fixing missing values, removing duplicates, etc
- Standardization: Enforcing consistent formats (e.g., changing dates to `YYYY-MM-DD` or converting all currencies to USD).
- Filtering: Selecting only the relevant columns and rows needed for business intelligence.
- Joining/Aggregation: Combining multiple data streams and summarizing data (e.g., calculating daily total sales).

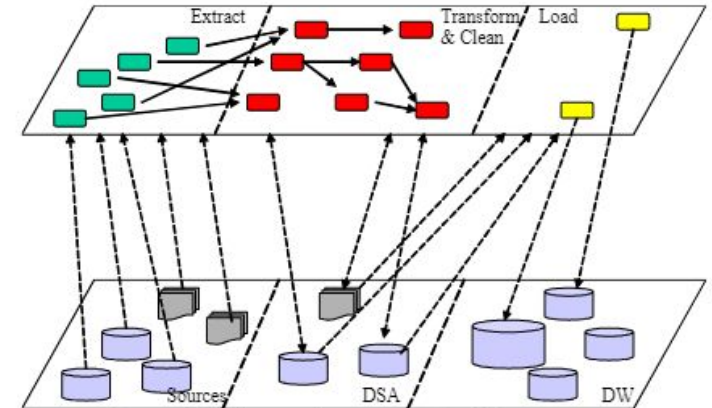


Figure 1. The environment of ETL processes

3.3 Load

Writing the transformed, clean data into the target destination.

Target Systems: Data Warehouses (Snowflake, Google BigQuery, Amazon Redshift) or Data Lakes.

Loading Strategies:

- Initial Load: Populating an empty table for the first time.
- Incremental Load: Adding new records into existing tables.
- Upsert (Update + Insert): Updating existing records if they have changed, or inserting them if they are completely new.

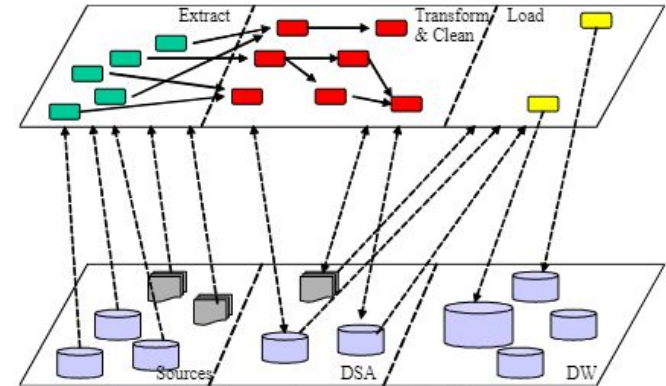


Figure 1. The environment of ETL processes

Example 3: ETL

- <https://github.com/ekaratnida/Data-Engineer/blob/main/de/ETL/lec1.py>

Get started

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Introduction

Prefect is an open-source orchestration engine that turns your Python functions into production-grade data pipelines with minimal friction. You can build and schedule workflows in pure Python—no DSLs or complex config files—and run them anywhere you can run Python. Prefect handles the heavy lifting for you out of the box: automatic state tracking, failure handling, real-time monitoring, and more.

ETL VS ELT

Feature	ETL (Traditional)	ELT (Modern Cloud)
Sequence	Transforms data before loading	Loads raw data first, then transforms
Processing Engine	Done by a separate ETL server (e.g., Spark).	Done by the Target Data Warehouse itself.
Data Flexibility	Only structured, clean data hits the warehouse.	Raw data is preserved, allowing future reimagining of transformations.